GRAPH ANALYTICS · FINAL PROJECT

Following the money behind Polymarket's Iran war bets

Winning wallets, money flows and suspicious patterns on markets resolved between January and April 2026

David and Wiyao · November 2026

Project objective

Find the biggest winners of Polymarket's Iran war markets and test whether some of them behave like one coordinated group.

01

Who won?

Profit of every wallet on the largest markets resolved in January–April 2026

02

Are winners linked?

Direct transfers and common funders in the money-flow graph

03

Where did gains go?

Transfers sent to related wallets after the markets resolved

We detect atypical on-chain patterns, not proof of insider trading.

Why these markets?

$397M

wagered on the five largest Iran markets we analysed

$2.4M

reportedly won by nine linked accounts with a 98% win rate on Iran bets

$1.2M

made by six wallets all funded in the 24 hours before the first US strikes

Public investigations describe a recurring pattern: **fund a new wallet, bet big at low odds, collect and move the gains**. We test it with graph analytics.

Sources: Bubblemaps analysis reported by CBS News (May 2026); Forbes (March 2, 2026); our data for the first figure.

How Polymarket works

EXAMPLE

US strikes Iran by February 28?



Buy 1,000 Yes shares for **$100**.

If Yes wins: **$1,000**. If No wins: $0.

Price = probability

Each share costs $0 to $1, and its price is the crowd's estimated chance of the outcome.

Peer to peer

Users trade with each other through an order book; there is no bookmaker taking the other side.

Resolution

An oracle settles the outcome by the market's written rules; winning shares pay $1 each.

Restricted access

Blocked in 30+ countries, including France since July 2026; some are close-only.

Close-only: existing positions can be sold or settled, but no new bets can be placed. Sources: Polymarket documentation; ANJ.

The blockchain in one slide

Block n

Deposit: exchange → 0x0a85

Bet: 0x7b02 buys Yes

→

Block n + 1

Transfer: 0xf70d → 0xbaa2

600,019 dollars in stablecoins

→

Block n + 2

Payout: winning shares → $1

Withdrawal: 0xbaa2 → 0xc011

Public ledger

Every transaction is recorded in blocks chained together: visible to all and impossible to rewrite.

Pseudonymous

Users are 0x... addresses, not names. Creating a new wallet takes seconds and costs nothing.

Polygon and stablecoins

Polymarket runs on Polygon; bets settle in dollar-pegged tokens (USDC.e, now pUSD).

Every transfer is an **edge waiting to be drawn**: this is what makes the money-flow graph possible.

Simplified illustration: wallets from our dataset, block contents not to scale.

The dataset: four public sources

Gamma API

Markets

Iran-tagged events, winning outcome, resolution time, volume

→

Data API v2

Trades

Every buy and sell: wallet, outcome, size, price, time

→

Activity feed

Flows

Deposits, withdrawals and tips between users

→

Polygon node

Who

Transaction receipts reveal the wallet on the other side

Polymarket runs on the Polygon blockchain, so every bet and every transfer is public. **The wallet address is the key that joins bets and money flows.**

gamma-api.polymarket.com · data-api.polymarket.com/v2 · public Polygon RPC nodes — no API key needed

Cleaning: what we kept

BETS

2,195
Iran markets found in 1,061 tagged events

1,131
paid out between Jan 1 and Apr 30, 2026

5
largest by volume, $397M wagered

926,658
trades by 82,596 distinct wallets

150
winners, ≥ $1,000 profit, $16.3M in total

MONEY FLOWS

500
wallets followed: winners and biggest stakers

55,258
deposits, withdrawals and tips

1,163
with a counterparty found on-chain

226
addresses, 380 transfer edges

220
nodes, 332 edges after removing mint/burn

Only markets that actually paid out are kept: voided or 50/50 markets have no winner. The filter uses the real resolution time, not the scheduled end date.

The five markets analysed

Market	Outcome	Resolved	Volume	Trades
Khamenei out as Supreme Leader of Iran by February 28?	Yes	Mar 4, 2026	$131.1M	200,000*
US strikes Iran by February 28, 2026?	Yes	Feb 28, 2026	$89.7M	200,000*
Khamenei out as Supreme Leader of Iran by March 31?	Yes	Mar 4, 2026	$63.2M	173,557
Will the Iranian regime fall by March 31?	No	Apr 1, 2026	$63.2M	200,000*
Khamenei out as Supreme Leader of Iran by January 31?	No	Feb 1, 2026	$49.7M	153,101

Three of the five markets resolved **Yes** within a week of the first strikes: the ones where a well-informed bet paid the most.

* Download cap reached: the API returns the most recent trades first, so the oldest trades of these markets are missing.

From transfers to a graph

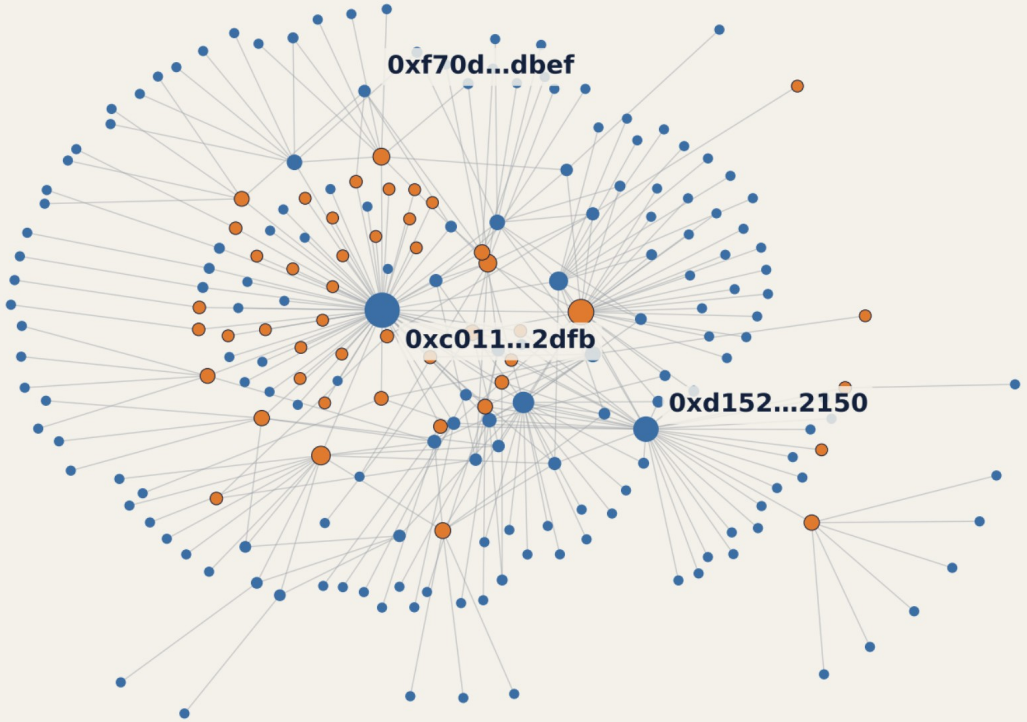
Nodes · 220

Wallet addresses, tagged with winner status and profit

Edges · 332

Directed A → B when A sent money to B; total amount, number of transfers, first and last date

48 winners other wallets



Structure and degree

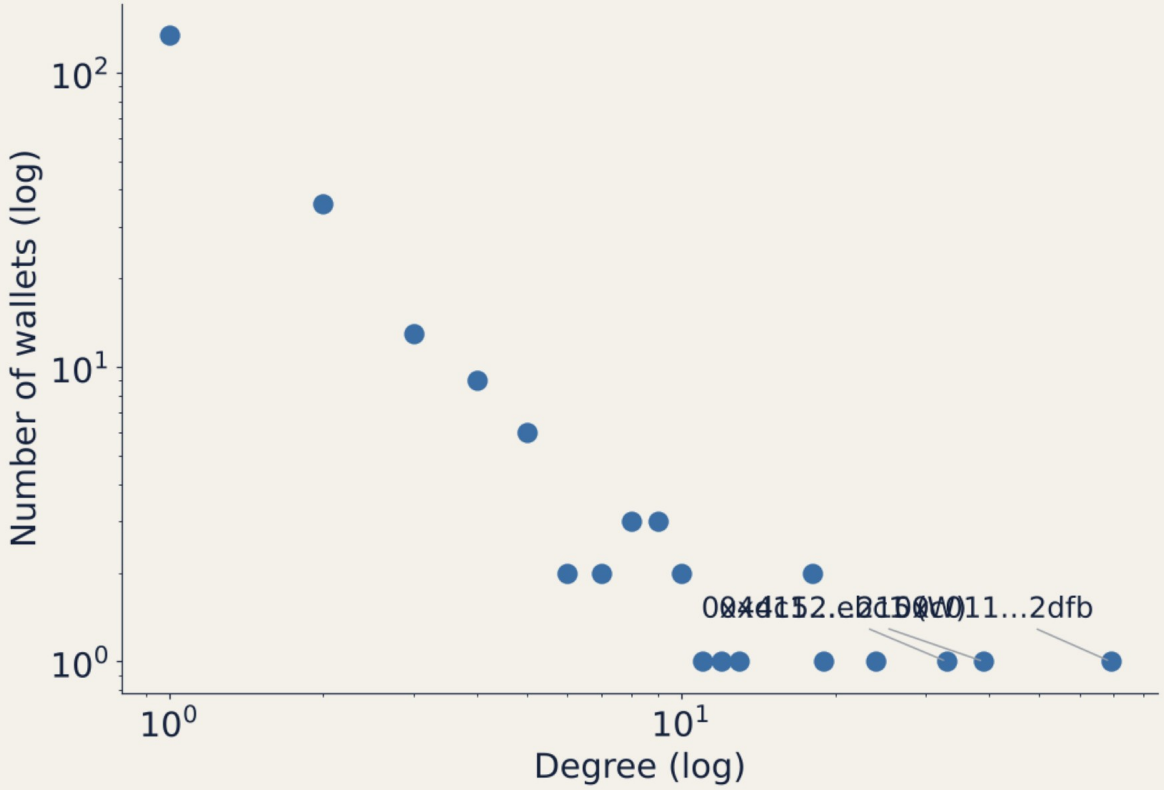
220
nodes (wallets)

332
directed, weighted edges

0.0069
density: share of possible edges

3.0
mean degree; 4.3 for winners, max 69

Extremely sparse and **heavy-tailed**: most wallets have one or two counterparties, a handful of hubs concentrate the links.

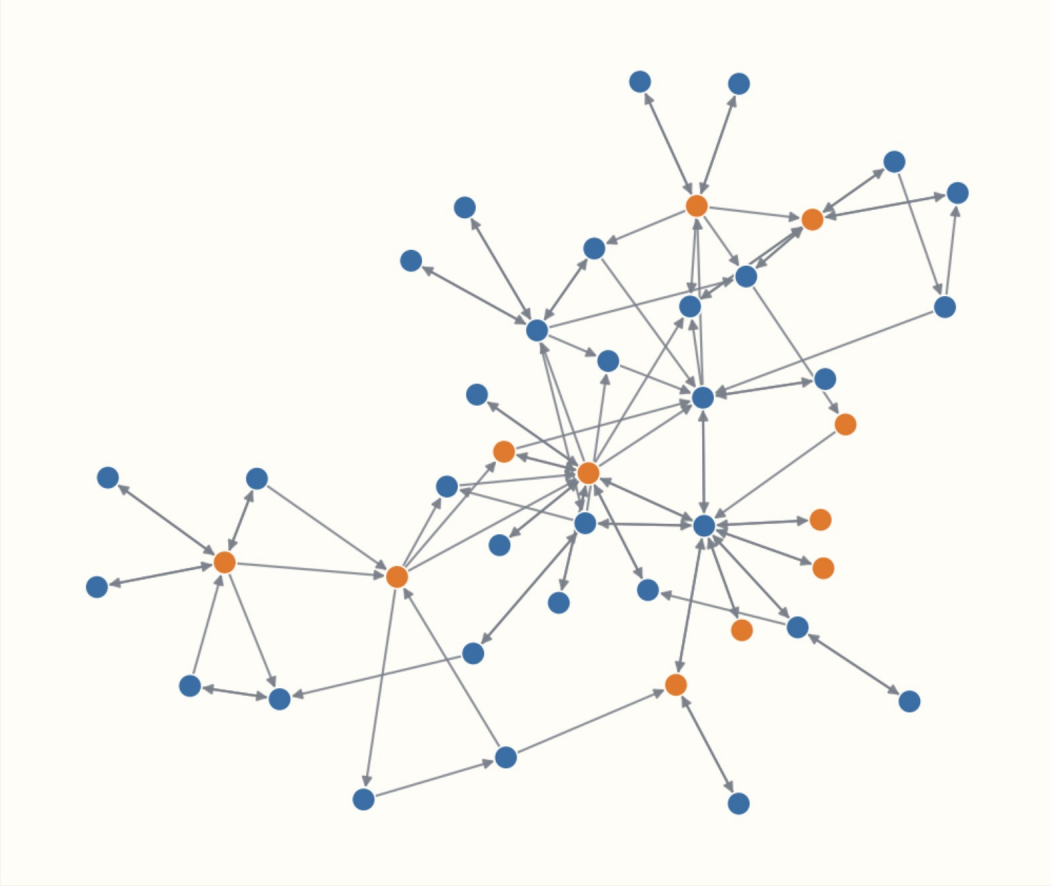


Connectivity and money loops

Connectivity

Weak components	**4**
Largest share	**97.3%**
Strong components	**170**
Money loops	**7**
Largest loop	**43**

Almost every wallet is reachable. In the largest loop, **money can travel from any wallet back to itself**: 43 wallets, 11 of them winners.



Distances, local structure, spectrum

The diameter path: six transfers between the two most distant wallets, through both main hubs



Distances

Diameter	**6**
Radius	**3**
Avg. path	**3.25**

Local

Clustering	**0.046**
Transitivity	**0.028**
Winners	**0.025**

Spectral

Fiedler λ_2	**0.068**

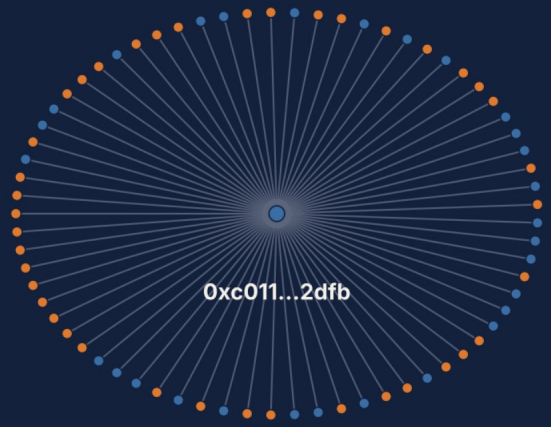
Normalized Laplacian, largest component

A **small world held together by hubs**: wallets are about three transfers apart, triangles are rare, and a low λ_2 means the graph is easy to cut into groups.

Centrality: who holds the network

Wallet	Role	Top 10s	In	Out
0xc011...2dfb	Sink	5 / 5	69	0
0x44c1...ebc1	Winner	5 / 5	21	18
0x7c3d...5c6b	Relay	5 / 5	13	11
0x35bb...009b	Relay	4 / 5	6	13
0xe25b...0f1b	Winner	4 / 5	5	7
0xd152...2150	Funder	3 / 5	0	33
0x614d...1546	Winner	3 / 5	9	9
0x6139...6b7a	Winner	3 / 5	5	8

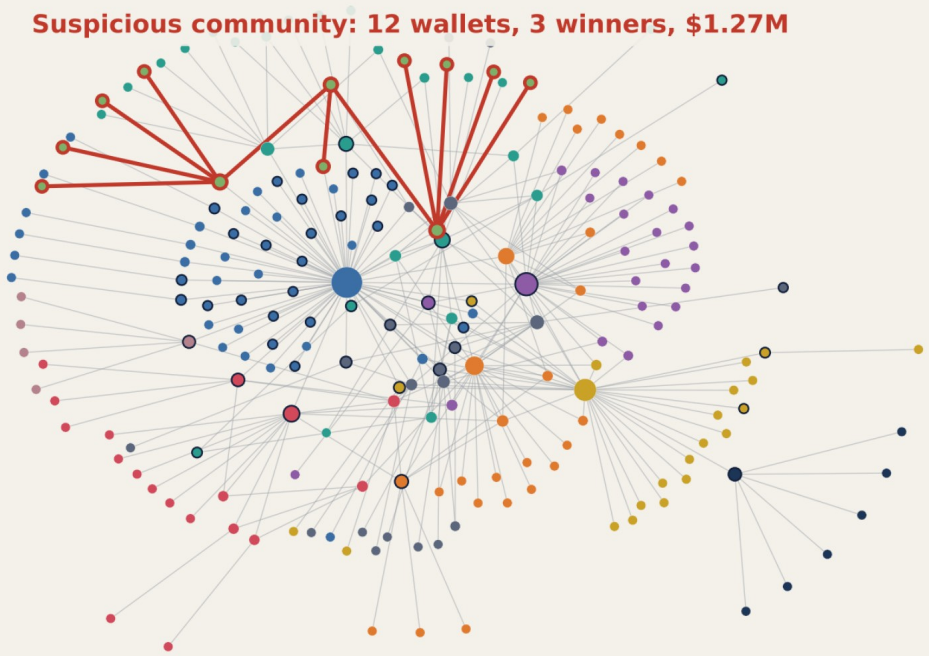
$26.8M



received by **0xc011...2dfb** from 69 wallets, 40 of them winners, and never sent on: an exit point, not a person.

Top 10s: number of top-10 lists among degree, betweenness, closeness, PageRank (by amount) and eigenvector centrality.

Communities (Louvain)



0.609 modularity

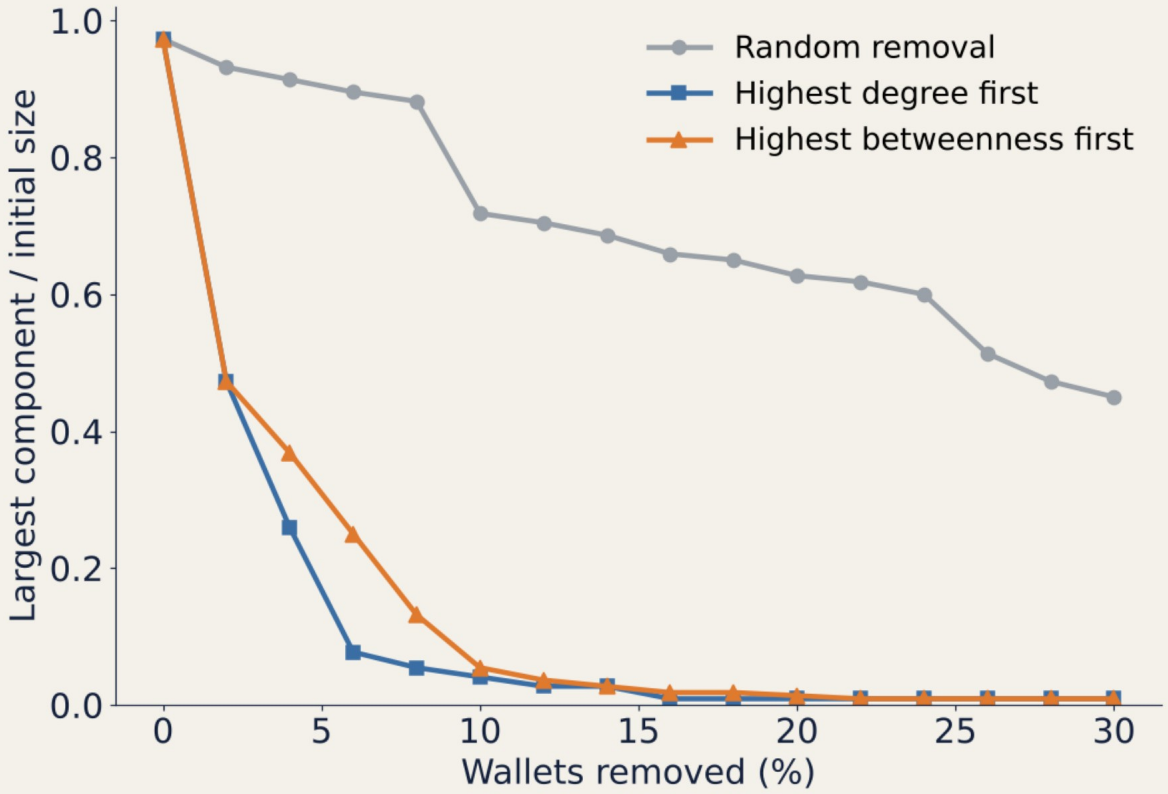
14 communities

Community	Wallets	Winners	Profit
#0 · 0xc011	51	23	$1.84M
#6 · 0x4cd0	12	3	$1.27M
#4 · 0x614d	23	3	$0.91M
#1 · 0x6139	27	5	$0.55M

Community #6 is small but earns **$1.27M with three winners**: we come back to it.

Undirected graph weighted by log(1 + amount). Each community is named after its most central wallet (PageRank).

Robustness: a network held by a few wallets



97% → 47%

Removing only **4 wallets** (the top 2% by betweenness) cuts the largest component in half.

Removing 10% of wallets at random still leaves 72% connected.

INSIGHTS

What sets some winners apart

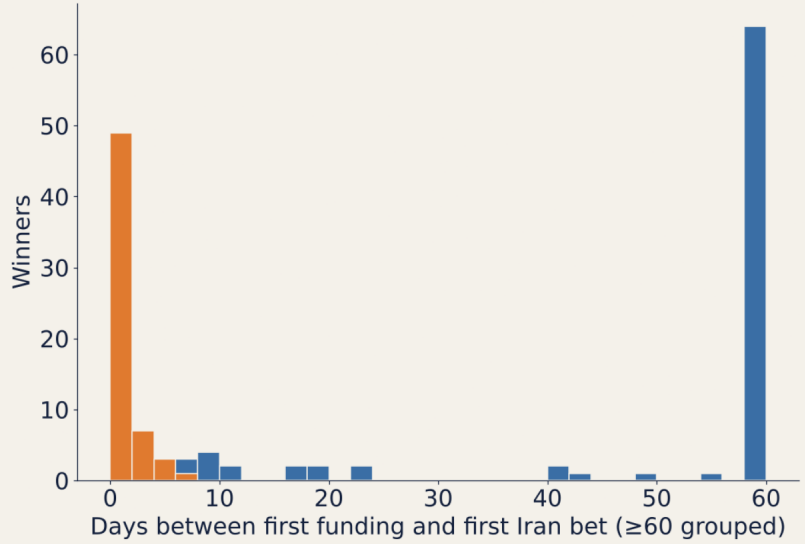
Fresh wallets

Long-shot wins

Linked winners

Redistribution of gains

Fresh wallets: funded, then straight to the bet



60
of 150 winners within 7 days

43
within 24 hours

20
within one hour

EXAMPLE · 0X7B02...4991

Funded on **Jan 29 at 13:54** UTC, first Iran bet at **14:06**, twelve minutes later.

$2.4M staked on 3 markets, all 3 won, 22 long-shot buys.

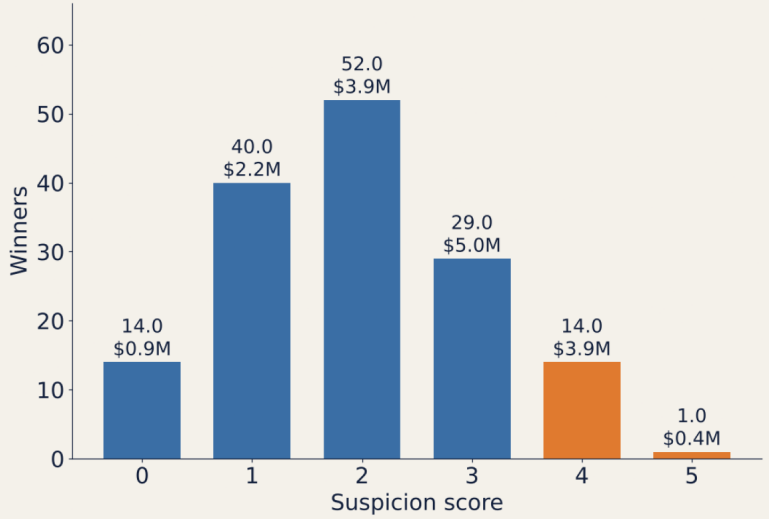
+$417k

Highest suspicion score in the dataset (5 of 6)

First funding = first deposit or incoming transfer seen in the wallet's Polymarket activity. 143 winners have a funding date.

Combining the signals: a suspicion score

Signal	Points
Fresh wallet (funded < 7 days before first bet)	1
At least one long-shot win (price < 0.35, stake ≥ $2.5k)	1
Two or more long-shot wins	+1
Win rate ≥ 80% on 2+ markets	1
Linked to another winner	1
Gains sent to a related wallet	1



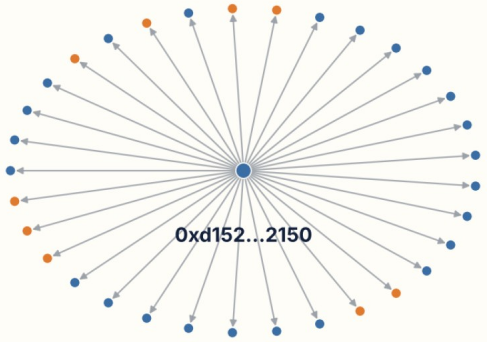
15 winners score 4 or more and earned $4.31M, 26% of all winners' profit.

Caveat: 92 of 150 winners had a long-shot win, because "Yes" was cheap before the events. A single signal is weak; the combination is what matters.

Linked winners: two clusters, two verdicts

CLUSTER A · WEAK

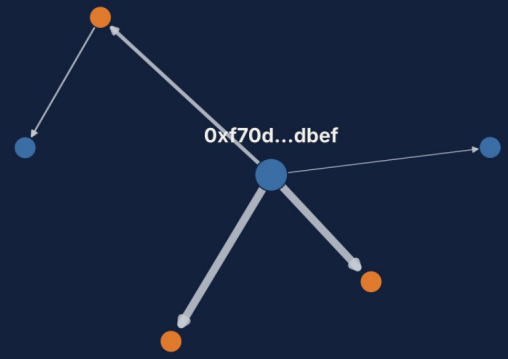
20 winners · $2.64M



0xd152...2150 sent $296k to 33 wallets, only $50k to winners: shared infrastructure, not a puppet master.

CLUSTER B · STRONG

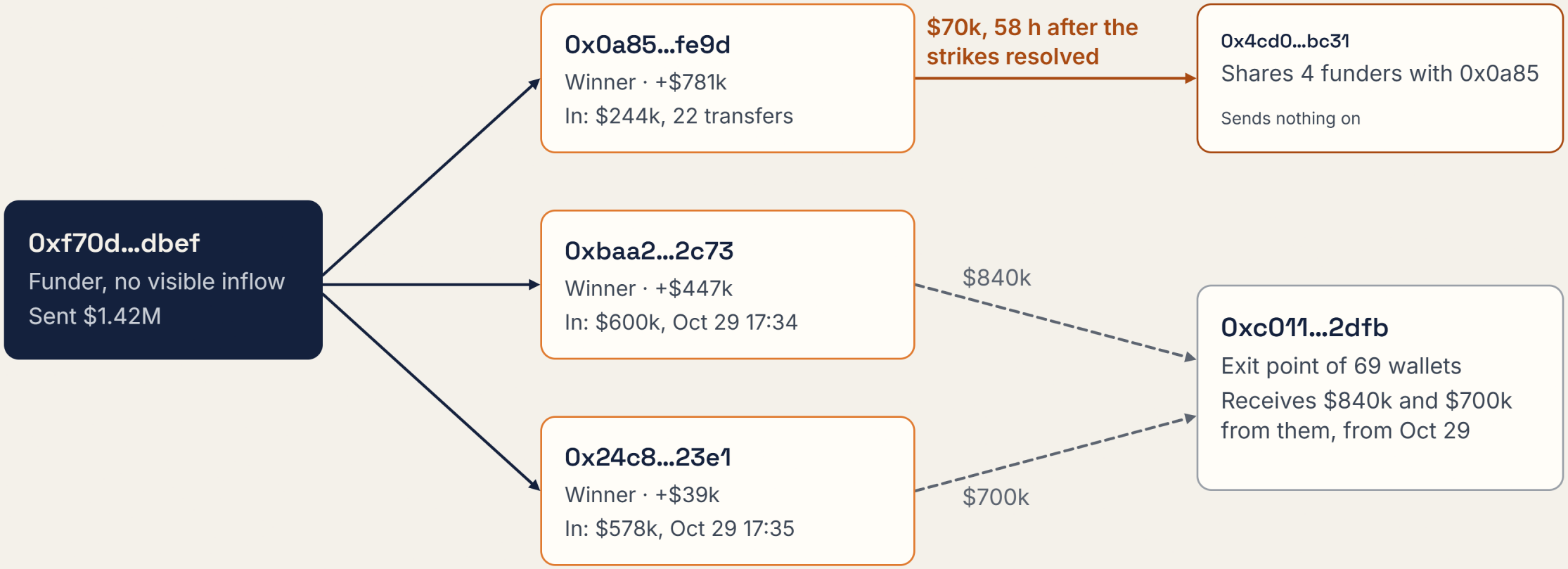
3 winners · $1.27M



0xf70d...dbef sent $1.42M to exactly these three winners: few, large, targeted transfers.

Linked = direct transfer or common funder within two hops; 23 of the 48 winners in the graph fall into a cluster. Winners in orange.

Following the money in cluster B



On Oct 29, within **25 seconds**, 0xf70d sends $1.18M to two winners and both send large sums to 0xc011: consistent with one operator.

Conclusion: what the clusters tell us

WHAT WE FOUND

One strong cluster

3 winners, $1.27M, one funder, synchronised flows

Fresh-wallet pattern

60 of 150 winners funded under a week before betting

Concentrated suspicion

15 high-score winners hold 26% of the profit

THEORY BEHIND IT

Informed trading

Kyle (1985): a trader with private information splits orders to hide it. Across wallets, the edge disappears from each account but shows in the graph.

Sybil wallets

One entity, many addresses: star motifs around a funder, high modularity (0.61) and a low algebraic connectivity ($\lambda_2 = 0.068$).

Hub-and-spoke flows

Heavy-tailed degrees, as in scale-free networks: a few funders and exits carry the money, so targeted removal breaks the graph.

Limits and next steps

LIMITS

Partial money trail

Only 1,163 of 55,258 flow rows (2%) were linked to a counterparty on-chain.

Five markets, capped trades

Three markets lost their oldest trades at the 200,000 cap.

Hub threshold too high

Exit points like 0xc011 stayed in and create false links.

Signals, not proof

Several wallets or shared funders can be legitimate.

NEXT STEPS

Full transfer history

Etherscan or Dune exports instead of activity rows.

Lower hub threshold

Test 20 to 50 and keep the clusters that stay stable.

Timing against the news

Compare bet times with the first public reports.

Temporal graph

Follow funding → bet → resolution → cash-out as a sequence.

Wallets are shown by shortened address only and are not linked to any real person.

Why it matters, and how to reuse it

USE CASES

Regulators

Rank accounts worth investigating; cases are real, as Israel's 2026 indictment of a reservist shows.

Platforms

Trigger identity checks or bet limits when winners cluster around one funder.

Journalists and researchers

Build open, verifiable evidence; measure how fast markets absorb private information.

Traders

Spot when they may be betting against better-informed money.

REPRODUCIBLE ON ANY TOPIC

Change four settings, rerun the notebook

TAG_SLUGS, KEYWORDS	the topic
START_DATE, END_DATE	the period
TOP_N_MARKETS	the markets kept
MIN_PROFIT	who counts as a winner

Elections · Central-bank rates · Sports transfers · Earnings and M&A · Awards · Product launches

Same pipeline every time: markets → trades → winners → money flows → graph → signals. Public data only.